UNMSM
Unidad de Posgrado de
Ingeniería de Sistemas e Informática

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
Doctoral Program in Deep Tech with a Focus on Artificial Intelligence and Emerging Technologies

DELIVERABLE No. 6 – REPRODUCIBILITY AUDIT REPORT

Research topic: Artificial intelligence governance maturity model for Peruvian organizations within the framework of Law No. 31814 and NTP-ISO/IEC 42001:2025: design, validation, and automated compliance assessment.

Audit type: Document-base preliminary audit of the published paper. Separate course-lab experiments were executed and verified in Google Colab; the published paper´s R repository hast not been independently executed. These are distinct evidence streams.

1. Paper metadata and rationale

Okyere, F., & Nyanney, M. (2026). Stability and interpretability of penalized logistic regression models for breast cancer risk prediction. PLOS ONE, 21(7), e0353489. DOI: https://doi.org/10.1371/journal.pone.0353489

Code repository: https://github.com/francisokyeregh/breast-cancer-penalized-logistic-regression

Dataset: Wisconsin Diagnostic Breast Cancer (UCI). The study compares unpenalized logistic regression, ridge, lasso, and elastic-net, emphasizing predictive performance, calibration, and bootstrap feature-selection stability. It was selected because the authors publicly identify the dataset and release R analysis code.

2. Experimental setup reported by the authors

Component	Evidence in published paper
Dataset	Wisconsin Diagnostic Breast Cancer; benign/malignant target; 30 predictors; source accessed 15 Dec 2025.
Data splitting	Stratified random 80% training / 20% held-out test.
Preprocessing	Continuous variables standardized using training-data parameters only.
Model selection	Penalized logistic regression tuned by cross-validation on training data.
Stability	Bootstrap resampling of training data; feature selection frequencies.
Evaluation	Accuracy, sensitivity, specificity, ROC AUC, Brier score, calibration; held-out test.
Reported headline	Penalized models: test accuracy 0.991; unpenalized: 0.965; elastic-net Brier score 0.012.

3. Reproducibility scorecard (Session 6, slide 40)

Criterion	Rating	Justification / evidence to verify
1. Seeds reported; many runs?	Partial	Bootstrap resampling is described, but a mean ± SD across independent training seeds is not reported as a headline result. Confirm seed controls in code.
2. Splits described?	Yes	Stratified 80/20 training/test split is explicitly described; exact row IDs should be checked in code.
3. Leakage checked?	Partial	Training-only scaling is present, but test labels are used to select Youden thresholds; optimized-threshold evaluation is not independent. Score remains Partial.
4. Tuning separated?	Yes	10-fold cv.glmnet tuning occurs on training data; separate test labels are later used for threshold optimization. Tuning criterion Yes is limited to model regularization.
5. Statistical test named?	No	A paired inferential test for comparing predictive models is not clearly identified in the inspected article sections.
6. Effect size and CI?	Partial	Performance differences are measurable, but confidence intervals for model differences are not clearly reported.
7. External validation?	No	Evaluation uses an internal held-out subset of one public dataset, not an independent hospital, time period, or population.
8. Compute documented?	Partial	R version, OS, libraries and execution environment were captured; no exact original package lockfile or hardware/runtime benchmark. Partial.
9. Code and data available?	Yes	Repository code and data were successfully retrieved and executed in Colab; archived commit, hashes, tables and figures. Yes.

Scoring rule: Yes = 1, Partial = 0.5, No = 0. Preliminary total = 5.0 / 9 (55.6%). This is an audit judgment, not a validated or official reproducibility index. Ratings may change after inspecting the repository and executing its code.

4. Data leakage and statistical rigor

The stated use of training-only standardization and held-out test set is a methodological strength. However, the audit must confirm that cross-validation preprocessing is refitted within each fold, bootstrap operations never touch the test set, and any Youden-threshold selection does not optimize on the final test labels. A single held-out split cannot establish cross-population generalizability. The reported 0.991 versus 0.965 accuracy difference is 2.6 percentage points, but a paired uncertainty estimate and statistical comparison are needed before claiming a robust superiority.

5. Independent reproduction protocol (executed and documented)

1. Record the repository URL, commit hash and access date; archive the original README and file inventory.
2. Obtain the exact authors' dataset and check dimensions, labels, missing values and checksums.
3. Install the declared R version and dependencies; record sessionInfo(), operating system and hardware.
4. Run the authors' original script unchanged; capture console output, execution time, errors and generated tables/figures.
5. Compare reproduced metrics against Table 1 and stable predictor counts against Table 2.
6. Repeat with documented seeds if feasible, recording mean ± SD and confidence intervals.
7. Log all discrepancies and any corrective modifications separately from the original run.

Outcome	Published	Independent reproduction (Colab)
Penalized model accuracy	0.991	0.9913 (optimized threshold)
Unpenalized accuracy	0.965	0.9652 (optimized threshold)
Elastic-net Brier score	0.012	0.01213
Stable predictors: ridge / lasso / elastic-net	30 / 9 / 22	30 / 9 / 22

6. Three highest-impact improvements

1. Publish a fully pinned execution environment (R version, package versions, lockfile), exact data version and a one-command reproduction procedure.
2. Report multi-seed performance, paired model-comparison uncertainty, confidence intervals and a clearly separated decision-threshold selection procedure.
3. Evaluate the models on a genuinely external dataset from a different institution, period or population, and report subgroup performance.

7. Overall judgement and relevance to doctoral research

The repository was independently executed with a path-only adjustment for Google Colab. The run completed and produced model objects, performance tables, and figures. This establishes computational executability in the tested environment, not external clinical validation or freedom from evaluation bias. The optimized classification thresholds were chosen on test labels; therefore threshold-dependent test-set metrics are potentially optimistic.

8. Exit ticket (Session 6, slide 51)

Questionable number: 99.1% test accuracy reported for penalized models. The number is not necessarily incorrect, but it comes from a single public benchmark with an internal held-out test set; it does not by itself establish performance in new hospitals or populations. It should be interpreted alongside uncertainty, external validation, calibration and reproducibility evidence.

References

- Okyere, F., & Nyanney, M. (2026). PLOS ONE, 21(7), e0353489. https://doi.org/10.1371/journal.pone.0353489
- Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science.
- Patterns, 4(9), 100804.
- Kapoor, S., et al. (2024). REFORMS: Consensus-based recommendations for machine-learning-based science. Science Advances, 10(18), eadk3452.
- Session 6: Experimental Best Practices (UNMSM), slides 39–40 and 51.

APPENDIX A — VERIFIED SESSION 6 LABORATORY RESULTS

Evidence source: Session6_Your_Turn_Completed (1).ipynb, downloaded after executing Runtime → Run all in Google Colab. The notebook contains 15 executed code cells with stored outputs and no recorded error outputs. The findings below are from course simulations, not from the audited PLOS ONE paper.

A1. Execution environment

Google Colab output: Python 3.13.16; Linux-6.6.122+-x86_64-with-glibc2.39; NumPy 2.1.3; scikit-learn 1.6.1; SciPy 1.16.3. The notebook is the primary evidence for the exact code, seeds, plots and execution sequence.

A2. Original six-lab execution checks

Lab	Recorded result	Interpretation
1 · Seed lottery	97.2% ± 1.0%; range 3.5 pp; 20 seeds	A single best run (98.2%) overstates typical performance.
2 · Leakage	Leaky 97.60%; clean 97.56%; pure-noise selection 90% vs clean 45%	Leakage can inflate apparent accuracy, dramatically on noise.
3 · Patient twins	Scan split 99%; patient split 54%	Patient-level grouping prevents identity overlap.
4 · Peeking / multiple testing	10 looks: 18.6% false winners; one look: 4.7%; multiple comparisons 66% vs Holm 4.7%	Repeated looks and comparisons inflate false positives.
5 · Statistical rigor	Tiny pilot p=0.202; d=0.41; 95% CI [-1.9, +8.8]; bootstrap accuracy 95.9% CI [93.0%, 98.8%]	Report uncertainty and effect magnitude.
6 · Fair comparison	Paired gap +1.64 pp; naive p≈0.0000; corrected p=0.0724	Overlapping CV folds make naive independent tests overconfident.

A3. Five completed "Your Turn" experiments

Task	Observed comparison	Conclusion
Lab 1: hidden layers (16,) → (64,)	Mean 97.193% → 97.105%; SD 1.015 → 0.919 pp; range unchanged 3.509 pp	Slightly lower SD, not a smaller observed range; larger networks do not automatically eliminate seed sensitivity.
Lab 2b: k = 5, 20, 100	Leaky / clean: 78/49%, 90/45%, 99/48%	Largest observed inflation at k=100: +51 pp.
Lab 3: fingerprint 3.0 → 0.5	By scan / patient: 99.25/54.25% → 56.25/48.75%	Gap decreases from 45 to 7.5 pp; group-aware evaluation remains essential.
Lab 4: 10 → 50 peeks	False positive rate 18.6% → 31.5%; single look 4.65%	Optional stopping inflates the nominal 5% error rate.
Lab 5b: n=20 → n=70 per group	n=20: p=0.20163, d=0.411, CI [-1.914, 8.776]; n=70: p≈0.000042, d=0.715, CI [3.407, 9.389]	Larger samples increase precision and power; the difference in observed d does not establish a change in the underlying effect.

A4. Reproducibility and integrity statement

The course-lab notebook demonstrates execution of six instructor labs and five Your Turn tasks. Separately, the authors' R analysis has now been independently executed and its generated evidence archived. These are distinct experimental streams.

APPENDIX B — EXIT TICKET

A number to question: the published paper's reported 99.1% accuracy on the Wisconsin breast-cancer benchmark. This figure is not necessarily incorrect; however, one internal holdout does not establish robustness across new institutions, populations, time periods or random seeds. I would request independent external validation, seed variability, confidence intervals and a leakage audit before making general deployment claims.

APPENDIX C — SUBMISSION CHECKLIST

- Main audit report: paper metadata, experimental setup, leakage, statistics, missing information, 9-item scorecard, score and three improvements.
- Executed Session 6 notebook (.ipynb) with stored Colab outputs, 15 executed code cells and no errors.
- This integrated report (DOCX/PDF), including verified lab results and Exit Ticket.
- Next research step: rerun threshold selection within training folds, evaluate once on untouched test data, and assess multi-seed uncertainty and external validation.

APPENDIX D — INDEPENDENT REPRODUCTION OF THE PUBLISHED STUDY

Evidence inspected: Deliverable6_Independent_Reproduction_Evidence.zip and Deliverable6_Independent_Reproduction_Colab(2).ipynb (uploaded 18 October 2026). The ZIP, not the notebook outputs, is the primary proof of successful execution: the uploaded notebook has no stored cell execution counts or output cells. The execution log ends with "DONE" and lists generated tables, figures, and saved model objects.

D1. Provenance and environment

- Git commit: 788f6280c2bc823d55885b0044ba18a564e24ef1
- Original R script SHA-256: 2eabc8510fad73825caef2ea124e51b7ebc671ac7b8405407ac352f2189679f9
- Dataset SHA-256: 1425d9affa78ba8e53afc81d0ef8a19069ee10c4b21fe89b3cf514071b12ee33
- Run metadata timestamp: 2026-10-18 05:11:26 UTC
- Execution environment: Google Colab; Linux Ubuntu 24.04.5 LTS; Python 3.13.16; R 4.6.1 (repository README requests R 4.5.0).
- The path-only adapted R script was executed. This is not an exact dependency-version replication.

D2. Independently generated metrics

Model	AUC	Accuracy (0.5)	Accuracy (Youden)	Brier score
Ridge	0.99935	98.26%	99.13%	0.02138
LASSO	0.99935	98.26%	99.13%	0.01470
Elastic Net	0.99935	98.26%	99.13%	0.01213
GLM	0.97158	95.65%	96.52%	0.04348

The 0.5-threshold results are preferable for descriptive holdout reporting because the Youden thresholds in the script are selected using y_test. The AUC and Brier scores do not depend on that threshold choice. The original code nevertheless has potential other sources of uncertainty that this run does not resolve.

D3. Bootstrap stability

Bootstrap feature-selection frequency: 1,000 bootstrap replicates; stability cutoff ≥0.70. Ridge: 30 stable features; LASSO: 9; Elastic Net: 22. Output tables and coefficient/ROC/calibration figures are included in the evidence archive.

D4. Integrity observations and limitations

- The R run completed with warnings: “glm.fit: algorithm did not converge” and “glm.fit: fitted probabilities numerically 0 or 1 occurred.” These require disclosure, particularly for the unpenalized logistic regression baseline.
- The script uses pROC::coords on y_test to choose optimal Youden thresholds for GLM, Ridge, LASSO, and Elastic Net, then reports performance on that same test set. This is test-set reuse and can inflate threshold-dependent metrics.
- The available execution is one seeded train/test partition (set.seed(42)); the 1,000 bootstrap replicates concern feature-selection stability, not independent multi-seed estimates of predictive performance.
- No external clinical cohort, confidence intervals for paired model differences, or prospective validation is established by this reproduction.
- The source repository README refers to a manuscript “under review”, while the previously cited published article metadata must be checked against the publisher for final bibliographic accuracy.

D5. Updated audit judgment

The original provisional nine-item score remains 5.0/9 under the same scoring rubric: independent execution resolves whether the shared code runs, but does not eliminate the missing multi-seed performance summaries, paired inferential uncertainty, external validation, or test-set threshold leakage. The evidence status is upgraded from “document-based only” to “independently executed with methodological caveats.”

Recommendation: for a stronger corrected analysis, select thresholds within cross-validation/training data only, retain an untouched test set for a single final evaluation, report fixed-threshold metrics and paired uncertainty, and repeat the full pipeline across seeds or independent cohorts.

D6. Evidence inventory

ZIP includes: audit_metadata.json; reproduction_run.log; results/sessionInfo.txt; results/tables/*.csv (metrics, stability, feature list); results/figures/* (ROC, calibration, coefficient paths, stability); results/rds/* (fitted models); original and path-adapted R scripts; data.csv; repository README and LICENSE.

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